M661 — Applied Statistics

Module 3 Assignment

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Math 661-854: Applied Statistics SP26 Online Virtual

Chapter 2: Looking at Data — Relationships

Tools used: Python 3 with pandas, numpy, scipy.stats, and matplotlib. All code and step-by-step calculations are shown throughout. Textbook: Introduction to the Practice of Statistics, 10th Edition (Moore, McCabe, Craig).

Task 1: Bone Strength — Exercises 2.14, 2.15, 2.16

Context: Osteoporosis is a condition where bones become weak, affecting over 200 million people worldwide. Exercise is one way to produce strong bones. Because we use our dominant arm more than our nondominant arm, we expect the bone in our dominant arm to be stronger. By comparing the strengths of both arms, we can get an idea of the effect that exercise has on bone strength. Data are measured in Newton meters divided by 1000 (Nm/1000) for 15 control subjects and 15 baseball players.

Exercise 2.14 — Bone Strength (Control Group)

Step 1: Load and examine the data

```
import pandas as pd
import numpy as np
from scipy import stats

df = pd.read_csv('ex02-014armstr.csv')
control = df[df['Group'] == 'Control']
```

Data for the 15 Control Subjects:

ID	NonDom (x)	Dom (y)	Difference (y – x)
1	15.7	16.3	0.6
2	25.2	26.9	1.7
3	17.9	18.7	0.8
4	19.1	22.0	2.9
5	12.0	14.8	2.8
6	20.0	19.8	-0.2
7	12.3	13.1	0.8
8	14.4	17.5	3.1
9	15.9	20.1	4.2
10	13.7	18.7	5.0
11	17.7	18.7	1.0
12	15.5	15.2	-0.3
13	14.4	16.2	1.8

14	14.1	15.0	0.9
15	12.3	12.9	0.6

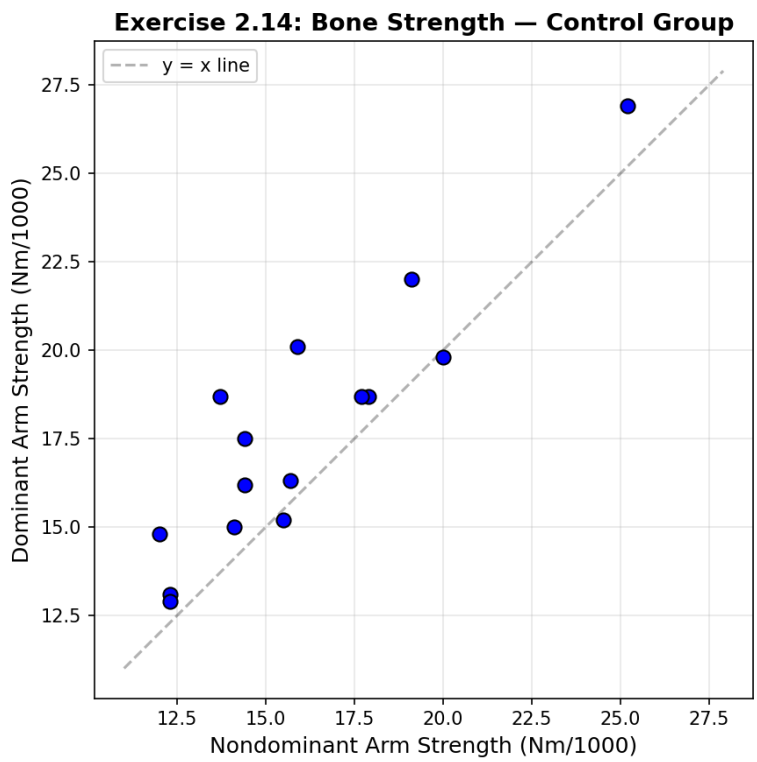
Step 2: Compute descriptive statistics

```
x_bar = control['NonDom'].mean()  # = 16.0133
y_bar = control['Dom'].mean()    # = 17.7267
sx = control['NonDom'].std(ddof=1) # = 3.5305
sy = control['Dom'].std(ddof=1)   # = 3.6517
r, p = stats.pearsonr(control['NonDom'], control['Dom'])
# r = 0.9049, p < 0.001
```

Results: The mean nondominant arm strength is x■ = 16.01 Nm/1000 (SD = 3.53), and the mean dominant arm strength is ■ = 17.73 Nm/1000 (SD = 3.65). The standard deviations are very similar, meaning both arms show roughly the same amount of variability in the control group. The mean dominant arm is about 1.71 Nm/1000 stronger.

Step 3: Create the scatterplot

```
plt.scatter(control['NonDom'], control['Dom'])
plt.xlabel('Nondominant Arm Strength (Nm/1000)')
plt.ylabel('Dominant Arm Strength (Nm/1000)')
plt.plot([11, 27], [11, 27], 'k--', label='y = x')  # reference line
```



Step 4: Describe the overall pattern and deviations

Overall pattern: The scatterplot shows a clear positive association between nondominant and dominant arm strength. As nondominant arm strength increases, dominant arm strength also increases. Most data points lie slightly above the dashed y = x reference line, which tells us that for most control subjects, the dominant arm is somewhat stronger than the nondominant arm — even without specialized exercise like throwing.

Deviations from pattern: Two subjects have a slightly weaker dominant arm: Subject 6 (NonDom = 20.0, Dom = 19.8, diff = –0.2) and Subject 12 (NonDom = 15.5, Dom = 15.2, diff = –0.3). These small negative

differences suggest that the dominant arm is not always stronger, but the deviations are very small. Subject 2 (NonDom = 25.2, Dom = 26.9) has the highest values but follows the overall trend. No point deviates dramatically from the overall pattern.

Step 5: Assess form, direction, and strength

Form: The relationship is approximately **linear**. The points follow a straight-line pattern with no evidence of curvature.

Direction: The relationship is **positive**. Individuals with stronger nondominant arms also tend to have stronger dominant arms. This makes biological sense — people with larger bone structure or more overall activity will have stronger bones in both arms.

Strength: The relationship is **strong**, with correlation r = 0.9049. The points cluster tightly around a line. Knowing nondominant arm strength gives a good prediction of dominant arm strength in the control group.

Step 6: Identify outliers

There are **no clear outliers**. All 15 points follow the general positive linear trend. Subject 2 has the highest values but is consistent with the pattern. Subjects 6 and 12 fall slightly below the y = x line, but neither deviates far enough from the overall pattern to be classified as an outlier.

Step 7: Assess linearity

Yes, the relationship is approximately linear. There is no curvature visible in the scatterplot, and the strong correlation of r = 0.9049 supports the appropriateness of a linear description.

What I learned: In the control group, both arms have similar strength (mean difference of only 1.71 Nm/1000), and the strong correlation (r = 0.905) confirms that arm strength is largely determined by overall bone structure and general activity rather than arm-specific exercise. The similar standard deviations (3.53 vs. 3.65) reinforce this — both arms show comparable variability.

Exercise 2.15 — Bone Strength (Baseball Players)

Step 1: Load and examine the data

```
baseball = df[df['Group'] == 'Baseball']
```

ID	NonDom (x)	Dom (y)	Difference (y – x)
16	17.0	19.3	2.3
17	16.9	19.0	2.1
18	17.7	25.2	7.5
19	21.2	37.7	16.5
20	21.0	40.3	19.3
21	14.6	20.8	6.2
22	31.5	36.9	5.4
23	14.9	21.2	6.3
24	15.1	19.4	4.3
25	13.5	20.4	6.9
26	13.6	17.1	3.5
27	20.3	26.5	6.2
28	17.3	30.3	13.0
29	14.6	17.4	2.8
30	22.6	35.0	12.4

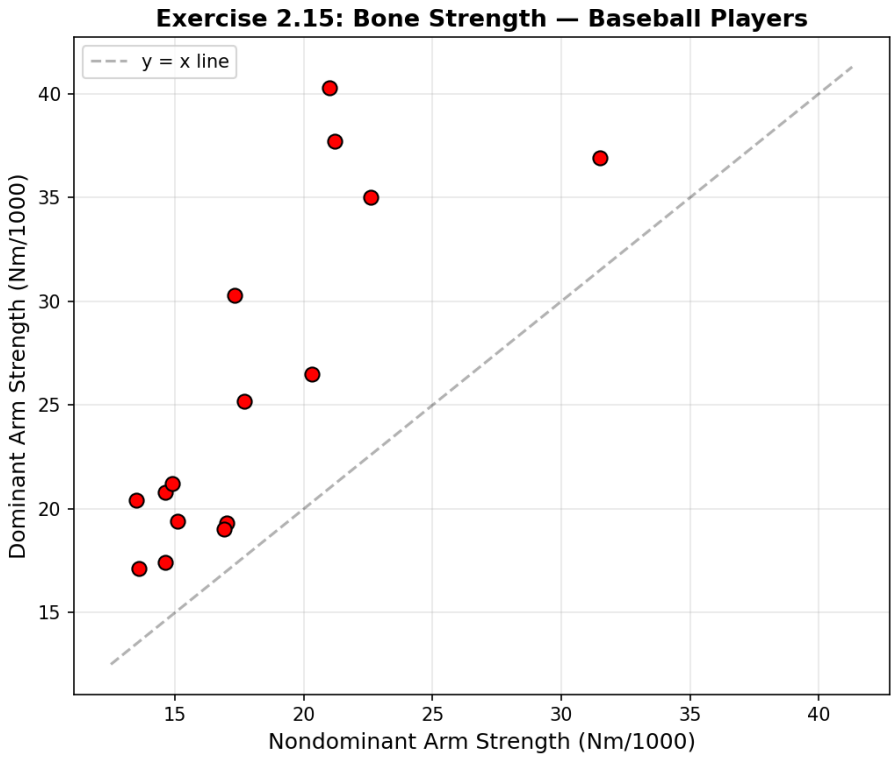
Step 2: Compute descriptive statistics

```
x_bar = baseball['NonDom'].mean()  # = 18.1200
y_bar = baseball['Dom'].mean()    # = 25.7667
sx = baseball['NonDom'].std(ddof=1) # = 4.7152
sy = baseball['Dom'].std(ddof=1)   # = 8.1583
r, p = stats.pearsonr(baseball['NonDom'], baseball['Dom'])
# r = 0.7936, p < 0.001
```

Results: Mean nondominant arm strength is x■ = 18.12 Nm/1000 (SD = 4.72), and mean dominant arm strength is ■ = 25.77 Nm/1000 (SD = 8.16). The standard deviations are very different — the dominant arm (SD = 8.16) is much more variable than the nondominant arm (SD = 4.72). This is because different baseball players throw with different intensity and frequency, creating variable amounts of bone strengthening. The mean dominant arm is 7.65 Nm/1000 stronger — much larger than the 1.71 gap in controls.

Step 3: Scatterplot

```
plt.scatter(baseball['NonDom'], baseball['Dom'], color='red', marker='^')
plt.plot(lims, lims, 'k--', label='y = x')
```



Step 4: Describe the overall pattern and deviations

Overall pattern: The scatterplot shows a positive association. The most striking feature is that **every single point lies above the y = x line** — every baseball player has a stronger dominant arm. The gap is large for most players, indicating the substantial effect of throwing exercise on dominant arm bone strength.

Deviations: The spread of dominant arm values is very large (17.1 to 40.3). Subjects 19 (diff = 16.5) and 20 (diff = 19.3) have exceptionally large gaps — their dominant arms are nearly twice as strong as their nondominant arms. Subject 22 (NonDom = 31.5) has the highest nondominant value, separated from the cluster along the x-axis.

Step 5: Assess form, direction, and strength

Form: Approximately **linear**, though with considerably wider scatter than controls.

Direction: Positive. Higher nondominant strength is associated with higher dominant strength.

Strength: Moderately strong, with r = 0.7936. Weaker than the control group (r = 0.9049) because throwing exercise introduces additional variability in dominant arm strength that cannot be predicted from nondominant arm strength alone.

Step 6: Identify outliers

Subject 20 (NonDom = 21.0, Dom = 40.3) has the highest dominant arm value and the largest arm difference (19.3). Subject 19 (NonDom = 21.2, Dom = 37.7) is also extreme. These could be potential outliers. Subject 22 (NonDom = 31.5) is potentially influential because its nondominant value is much higher than the rest of the data.

Step 7: Assess linearity

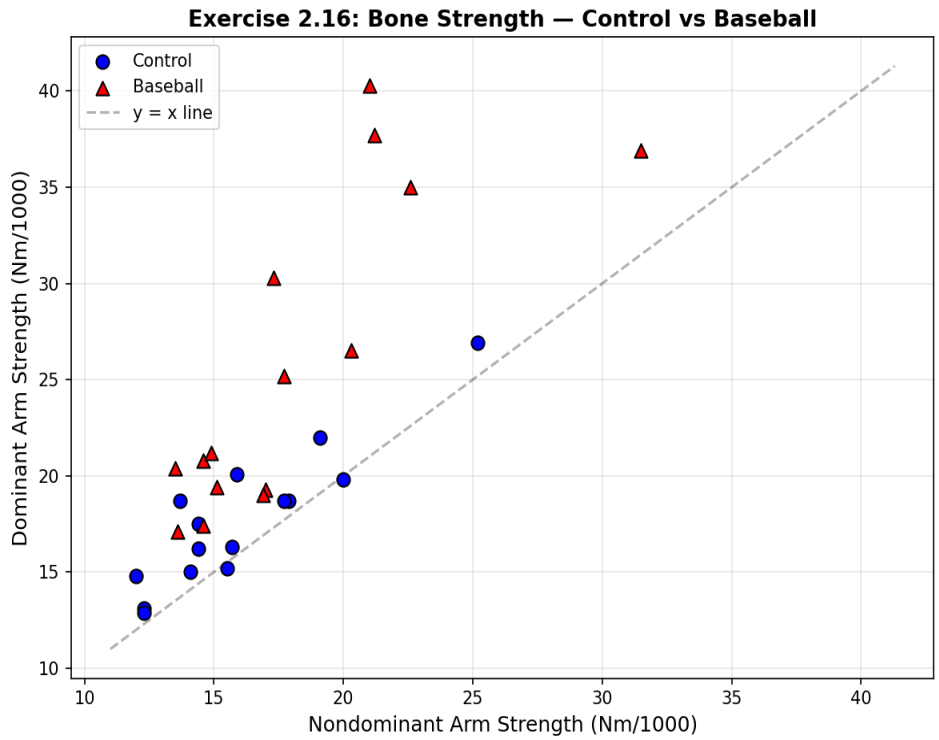
The relationship is approximately linear with no strong evidence of curvature. A linear model is reasonable, though the wider scatter means it will be less precise.

What I learned: Baseball players show a dramatically different pattern from controls. Every player's dominant arm is stronger (mean diff = 7.65 vs. 1.71 for controls), and the variability in dominant arm strength (SD = 8.16) is far greater than nondominant (SD = 4.72). This asymmetry is a clear signature of the throwing exercise effect.

Exercise 2.16 — Compare Baseball Players with Controls

Step 1: Create a combined scatterplot with different symbols

```
plt.scatter(control['NonDom'], control['Dom'], color='blue',
    marker='o', label='Control')
plt.scatter(baseball['NonDom'], baseball['Dom'], color='red',
    marker='^', label='Baseball')
plt.plot(lims, lims, 'k--', label='y = x')
```



Step 2: Describe and compare the relationships

Statistic	Control (n=15)	Baseball (n=15)
Mean NonDom (Nm/1000)	16.01	18.12
SD NonDom	3.53	4.72
Mean Dom (Nm/1000)	17.73	25.77
SD Dom	3.65	8.16
Mean Diff (Dom – NonDom)	1.71	7.65
Correlation r	0.9049	0.7936

Observation 1 — Dominant arm strength is much higher for baseball players: In the combined plot, the red triangles (baseball) are shifted upward compared to the blue circles (controls). Mean dominant arm strength is 25.77 for baseball vs. 17.73 for controls — a 45% increase. This is consistent with the hypothesis that throwing exercise strengthens the dominant arm's bones.

Observation 2 — The arm gap is much larger for baseball players: The mean Dom – NonDom difference is 7.65 for baseball vs. only 1.71 for controls — about 4.5 times larger. Baseball players' points are displaced far above the y = x line, while controls cluster near it.

Observation 3 — Nondominant arms are similar across groups: Mean nondominant strength is 18.12 (baseball) vs. 16.01 (controls) — a modest difference. The nondominant arm does not receive throwing exercise, so the difference between groups is concentrated in the arm that gets the extra exercise. This serves as an internal control.

Observation 4 — More variability in baseball group: SD of dominant arm: 8.16 (baseball) vs. 3.65 (controls). Individual differences in throwing intensity lead to a wider range of dominant arm strength. The weaker correlation (r = 0.794 vs. 0.905) reflects this added variability.

Observation 5 — Both groups show positive linear relationships: In both groups, people with stronger nondominant arms tend to have stronger dominant arms. The linear form is maintained in both, but the baseball group is shifted upward and more spread out.

Summary — What I learned: The combined plot powerfully illustrates the effect of exercise on bone strength. The two groups share similar nondominant arm strength (the 'baseline'), but the baseball players' dominant arms are dramatically stronger. The fact that the difference is isolated to the exercised arm — while the non-exercised arm is comparable — provides compelling evidence that throwing exercise specifically causes the increased bone strength. Comparing two arms within the same person controls for confounding variables like genetics, nutrition, and overall activity level.

Task 2: Bone Strength (Continued) — Exercises 2.35, 2.36

Exercise 2.35 — Correlation: Control Group

> **Question:** Find the correlation between the bone strength of the dominant arm and the nondominant arm for the 15 control subjects. Is the correlation a good numerical summary of the scatterplot? Explain.

(a) Find the correlation

The correlation r measures the direction and strength of the **linear** relationship between two quantitative variables. The formula is:

$$r = (1/(n-1)) \times \Sigma \left[((x_i - x\blacksquare)/s_x) \times ((y_i - \blacksquare)/s_y) \right]$$

Step 1: Compute the means and standard deviations

```
x = control['NonDom']  # nondominant arm strength
y = control['Dom']    # dominant arm strength
n = 15

x_bar = sum(x) / n = 240.2 / 15 = 16.0133
y_bar = sum(y) / n = 265.9 / 15 = 17.7267

sx = sqrt( sum((x_i - x_bar)**2) / (n-1) )
   = sqrt(174.4973 / 14) = sqrt(12.4641) = 3.5305
sy = sqrt( sum((y_i - y_bar)**2) / (n-1) )
   = sqrt(186.6893 / 14) = sqrt(13.3350) = 3.6517
```

Step 2: Compute the standardized values and their products

For each subject, we standardize both x and y, then multiply the z-scores:

ID	x	y	z_x = (x–x■)/s_x	z_y = (y–■)/s_y	z_x × z_y
1	15.7	16.3	-0.0887	-0.3907	+0.0347
2	25.2	26.9	+2.6021	+2.5121	+6.5366
3	17.9	18.7	+0.5344	+0.2665	+0.1424
4	19.1	22.0	+0.8743	+1.1702	+1.0231
5	12.0	14.8	-1.1368	-0.8015	+0.9111
6	20.0	19.8	+1.1292	+0.5678	+0.6411
7	12.3	13.1	-1.0518	-1.2670	+1.3326
8	14.4	17.5	-0.4570	-0.0621	+0.0284
9	15.9	20.1	-0.0321	+0.6499	-0.0209
10	13.7	18.7	-0.6552	+0.2665	-0.1746
11	17.7	18.7	+0.4778	+0.2665	+0.1273
12	15.5	15.2	-0.1454	-0.6919	+0.1006
13	14.4	16.2	-0.4570	-0.4181	+0.1910
14	14.1	15.0	-0.5419	-0.7467	+0.4047
15	12.3	12.9	-1.0518	-1.3218	+1.3902

Step 3: Compute the correlation

```
r = (1/(n–1)) × Σ(z_x × z_y) = (1/14) × 12.6684 = 0.9049
```

Verification using software:

```
r, p_value = stats.pearsonr(control['NonDom'], control['Dom'])
# r = 0.9049, p_value = 3.68e-06
```

The correlation is r = 0.9049.

(b) Is the correlation a good numerical summary? Explain.

Yes, the correlation is a good numerical summary of this scatterplot. Here is why:

1. The relationship is approximately linear. Correlation measures the strength and direction of a *linear* relationship. Looking at the scatterplot from Exercise 2.14, the points follow a straight-line pattern with no evidence of curvature. Because the relationship is linear, the correlation captures the essential feature of the association.

2. There are no extreme outliers distorting the value. Correlation is sensitive to outliers — a single extreme point can dramatically change the value of r. In this dataset, all 15 points follow the general trend. No point is far removed from the overall linear pattern. Therefore, the correlation is not being distorted by unusual observations.

3. The value of r = 0.9049 accurately reflects the strong positive trend. Looking at the scatterplot, the points cluster fairly tightly around an upward-sloping line. A correlation of 0.90 correctly communicates this: it tells us that the relationship is both positive (as one arm gets stronger, so does the other) and strong (the points are not widely scattered). This matches the visual impression from the scatterplot.

4. The variables are appropriate for correlation. Both nondominant and dominant arm strength are quantitative variables measured on the same cases (the 15 control subjects), which are the requirements for using correlation.

In summary, because the relationship is linear, there are no problematic outliers, and both variables are quantitative, the correlation r = 0.9049 provides an excellent numerical summary that complements the graphical display of the scatterplot.

> **What I learned:** The correlation is a useful summary *only when* the relationship is approximately linear and free of extreme outliers. For the control group, both conditions are met. In cases where the relationship is curved (like calcium retention vs. intake) or dominated by an outlier (like the laundry detergent data with the expensive outlier), the correlation would not be a good summary — we would need to describe the relationship differently.

Exercise 2.36 — Correlation: Baseball Players

Question: Find the correlation between dominant and nondominant arm bone strength for the 15 baseball players. Is the correlation a good numerical summary of the scatterplot? Explain.

(a) Find the correlation

Step 1: Compute the means and standard deviations

```
x = baseball['NonDom']  # nondominant arm strength
y = baseball['Dom']    # dominant arm strength
n = 15

x_bar = sum(x) / n = 271.8 / 15 = 18.1200
y_bar = sum(y) / n = 386.5 / 15 = 25.7667

sx = sqrt( sum((x_i - x_bar)**2) / (n-1) )
   = sqrt(311.2640 / 14) = sqrt(22.2331) = 4.7152
sy = sqrt( sum((y_i - y_bar)**2) / (n-1) )
   = sqrt(931.8133 / 14) = sqrt(66.5581) = 8.1583
```

Step 2: Compute the standardized values and their products

ID	x	y	z_x = (x–x■)/s_x	z_y = (y–■)/s_y	z_x × z_y
16	17.0	19.3	-0.2375	-0.7927	+0.1883
17	16.9	19.0	-0.2587	-0.8294	+0.2146
18	17.7	25.2	-0.0891	-0.0695	+0.0062
19	21.2	37.7	+0.6532	+1.4627	+0.9555
20	21.0	40.3	+0.6108	+1.7814	+1.0881
21	14.6	20.8	-0.7465	-0.6088	+0.4545
22	31.5	36.9	+2.8376	+1.3647	+3.8724
23	14.9	21.2	-0.6829	-0.5598	+0.3823
24	15.1	19.4	-0.6405	-0.7804	+0.4998
25	13.5	20.4	-0.9798	-0.6578	+0.6445
26	13.6	17.1	-0.9586	-1.0623	+1.0183
27	20.3	26.5	+0.4623	+0.0899	+0.0416
28	17.3	30.3	-0.1739	+0.5557	-0.0966
29	14.6	17.4	-0.7465	-1.0255	+0.7656
30	22.6	35.0	+0.9501	+1.1318	+1.0753

Step 3: Compute the correlation

```
r = (1/(n–1)) × Σ(z_x × z_y) = (1/14) × 11.1103 = 0.7936
```

Verification using software:

```
r, p_value = stats.pearsonr(baseball['NonDom'], baseball['Dom'])
# r = 0.7936, p_value = 3.97e-04
```

The correlation is r = 0.7936.

(b) Is the correlation a good numerical summary? Explain.

The correlation is a reasonable numerical summary, but with some caveats.

1. The relationship is approximately linear. The scatterplot from Exercise 2.15 shows a generally linear trend — no clear curvature. This is a necessary condition for the correlation to be a useful summary, and it

is met here.

2. There are potential outliers and influential points that may affect r. Subject 22 (NonDom = 31.5, Dom = 36.9) has a nondominant arm strength value much higher than any other subject (the next highest is 22.6). This potentially influential observation could be pulling the correlation in one direction. Additionally, Subjects 19 and 20 have dominant arm values (37.7 and 40.3) that are far above the rest of the data. Correlation is sensitive to extreme values, so these points may be affecting the computed value of r. It would be wise to compute r with and without these points to assess their influence.

3. The scatter is substantially wider than in the control group. The correlation of r = 0.7936 is lower than the control group's r = 0.9049. This accurately reflects the wider scatter visible in the baseball scatterplot. The throwing exercise creates additional variability in dominant arm strength, which the correlation correctly captures as a weaker (but still moderately strong) linear relationship.

4. The value r = 0.7936 provides useful but incomplete information. While r tells us the direction (positive) and overall strength (moderately strong) of the linear relationship, it does not capture the most important feature of this dataset: that every baseball player's dominant arm is stronger than the nondominant arm, and the gap is large. The correlation alone does not convey this systematic shift above the y = x line. The scatterplot provides essential context that r alone cannot.

In summary, r = 0.7936 is a useful but somewhat limited summary for the baseball players. The linear form is approximately satisfied, making the correlation valid. However, the presence of potential influential points (Subject 22) and the fact that the most interesting feature (the systematic dominance gap) is not captured by correlation alone mean that the scatterplot is an essential complement to the numerical summary.

Comparison of correlations:

	Control	Baseball
Correlation r	0.9049	0.7936
Interpretation	Strong positive linear	Moderately strong positive linear
Good summary?	Yes — linear, no outliers	Reasonable — linear, but influential points present

What I learned: Correlation is a powerful summary when conditions are right (linear relationship, no extreme outliers), as in the control group. For the baseball players, r is still useful but incomplete — it captures the linear trend but misses the systematic shift caused by throwing exercise. This exercise reinforced an important lesson: **always examine the scatterplot alongside the correlation**. A numerical summary without a graphical display can be misleading, and vice versa. Together, they provide a complete picture of the relationship.